Decision trees and Ensembles

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Recap

Lecture 4:
SVM, PCA

1. Support Vector Machine (SVM)
   - Hinge loss
   - Kernel trick
2. Dimensionality reduction and PCA
   - Problem statement
   - Singular Value Decomposition
   - Eckart–Young theorem
   - Equivalent definitions
   - Data normalization

Outline

1. Intuition
2. Construction procedure
3. Information criteria
4. Special highlights
    - Dealing with missing data
    - Binarization
    - Decision tree as linear model
    - Standards
    - Hyperparameters
5. Bootstrap and Bagging
6. Random Forest

Decision Tree: intuition

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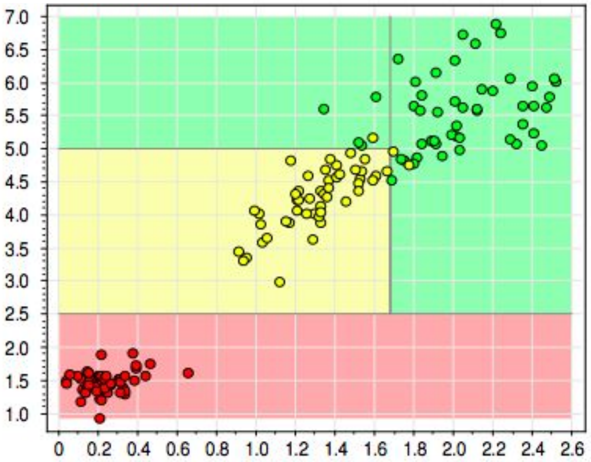
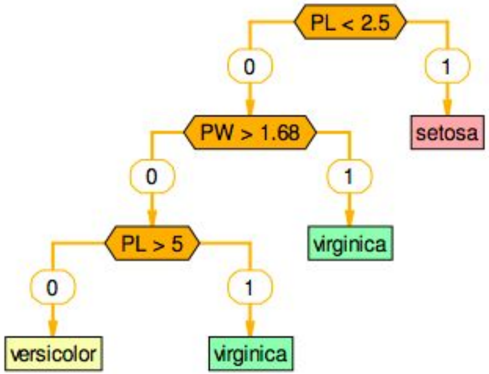
01

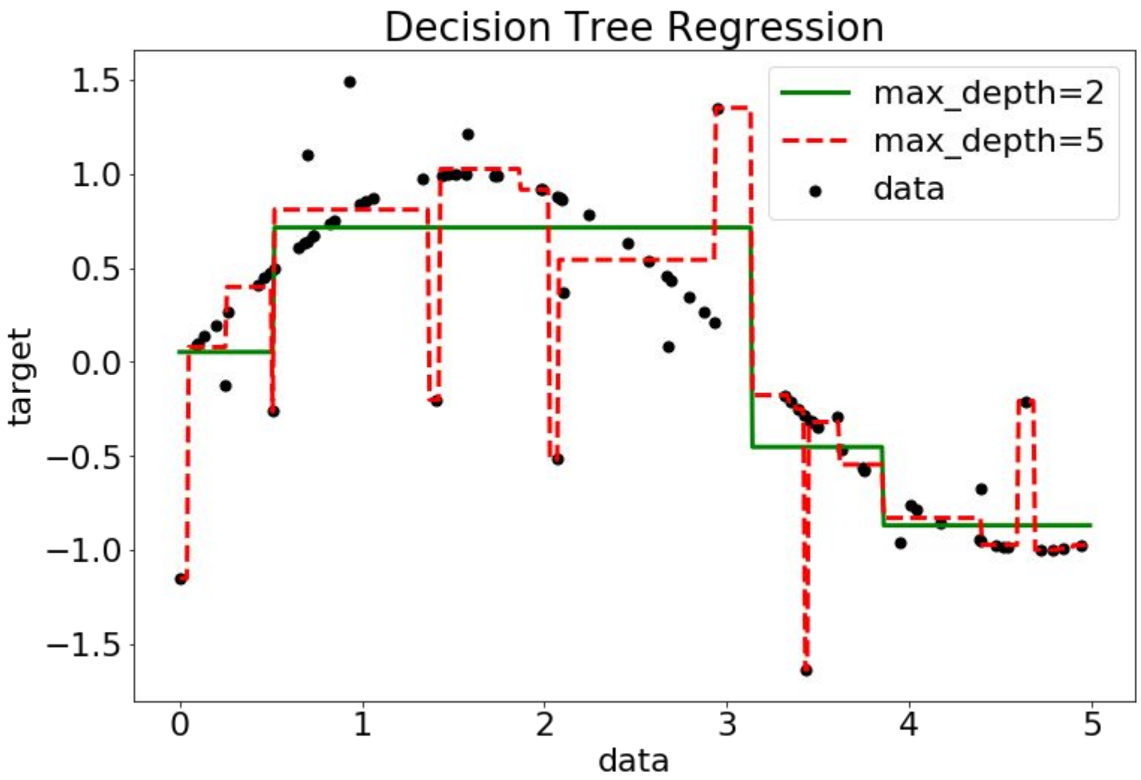
Decision tree for Firsher's Irises





setosa	$r_1(x) = [PL \leqslant 2.5]$
virginica	$r_2(x) = [PL > 2.5] \wedge [PW > 1.68]$
virginica	$r_3(x) = [PL > 5] \wedge [PW \leqslant 1.68]$
versicolor	$r_4(x) = [PL > 2.5] \wedge [PL \leqslant 5] \wedge [PW < 1.68]$

Decision tree in regression

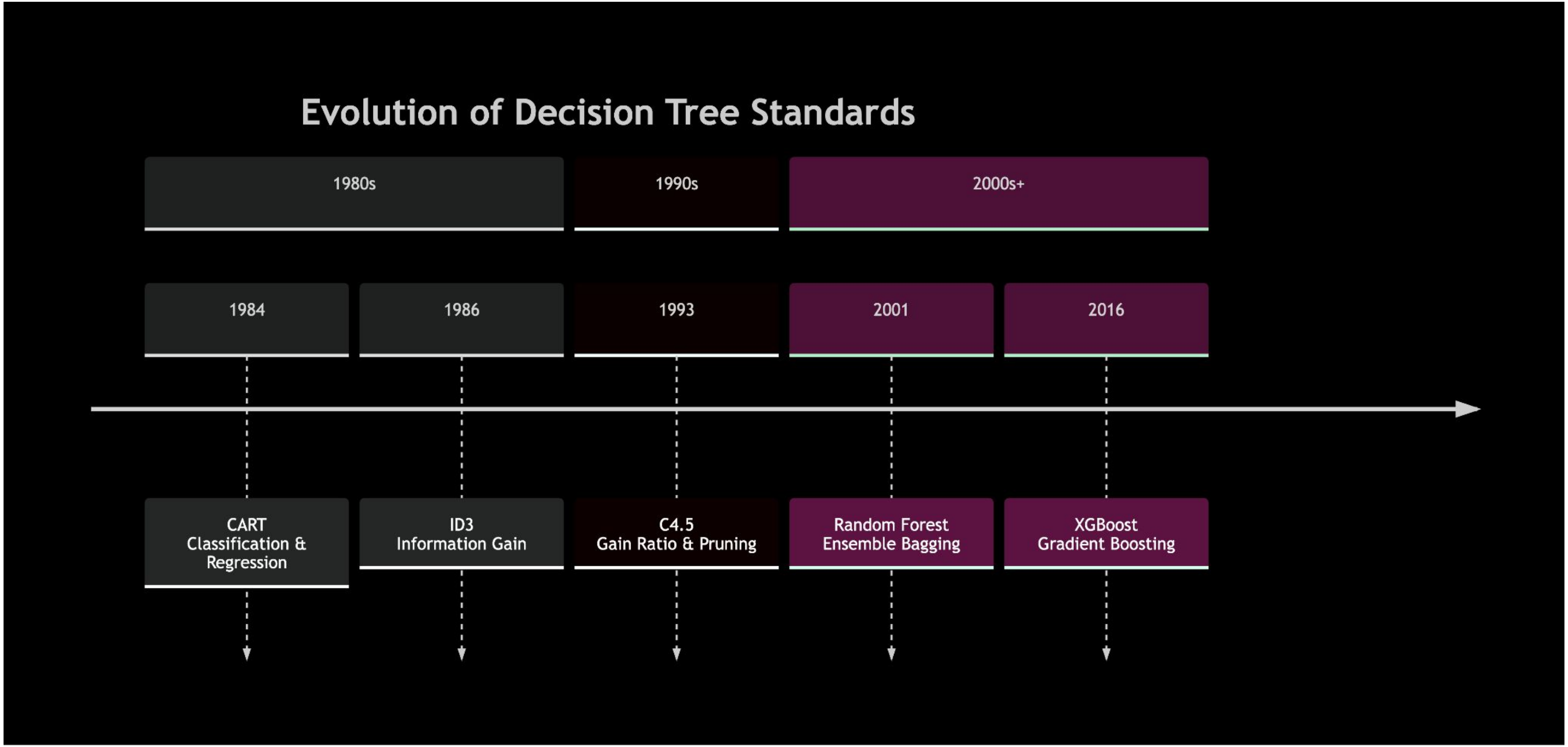


Green - decision tree of depth 2

Red - decision tree of depth 5

Every leaf corresponds to some constant.

Timeline of tree algorithms



Standards

- Iterative Dichotomiser 3 (ID3) (1986)
  - Entropy criteria; Stops when no more gain available
- C4.5 (1993)
  - Normalised entropy criteria; Stops depending on leaf size; Incorporates pruning
- ID5 (1994)
  - Introduces iterative learning enabling tree update after training
- CART (1984)
  - Gini criteria; Cost-complexity Pruning; Surrogate predicates for missing data;
  - Origin for most current implementations (sklearn, boosting libs)
- etc.

Read more

Decision Tree construction procedure

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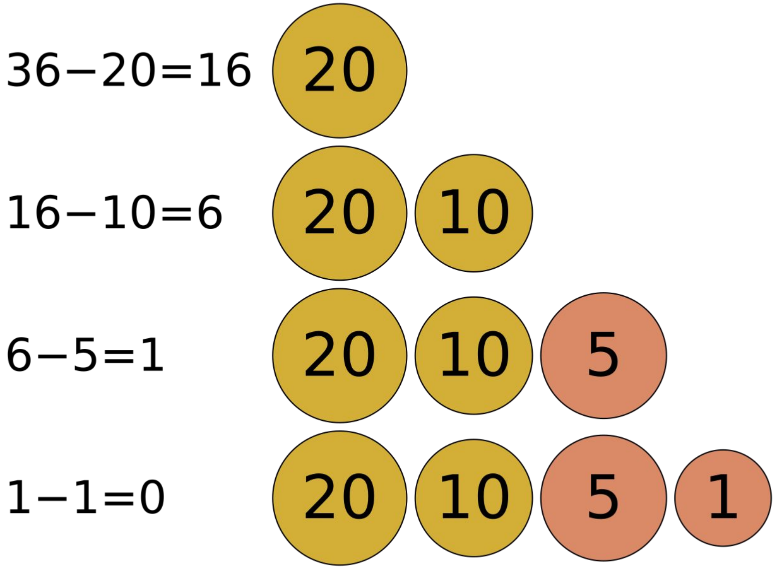
02

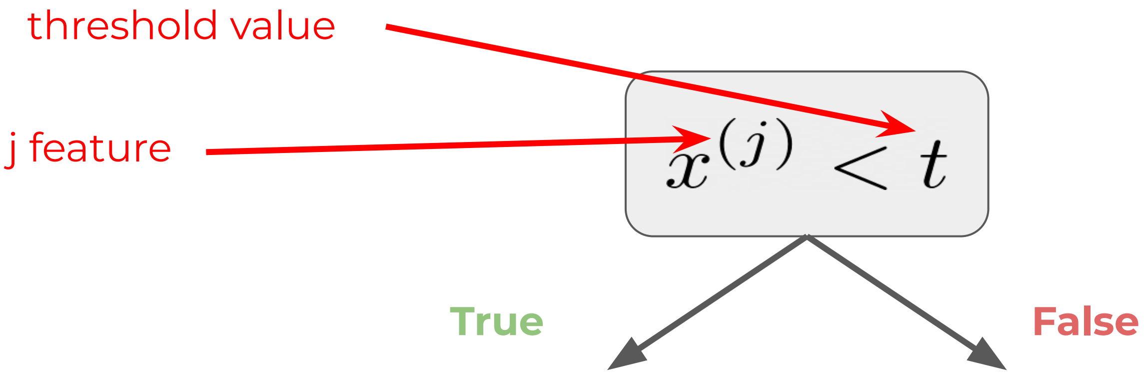
Greedy algorithms

A greedy algorithm is any algorithm that follows the problem-solving heuristic of making the locally optimal choice at each stage.

In many problems, a greedy strategy does not produce an optimal solution, but a greedy heuristic can yield locally optimal solutions that approximate a globally optimal solution in a reasonable amount of time.

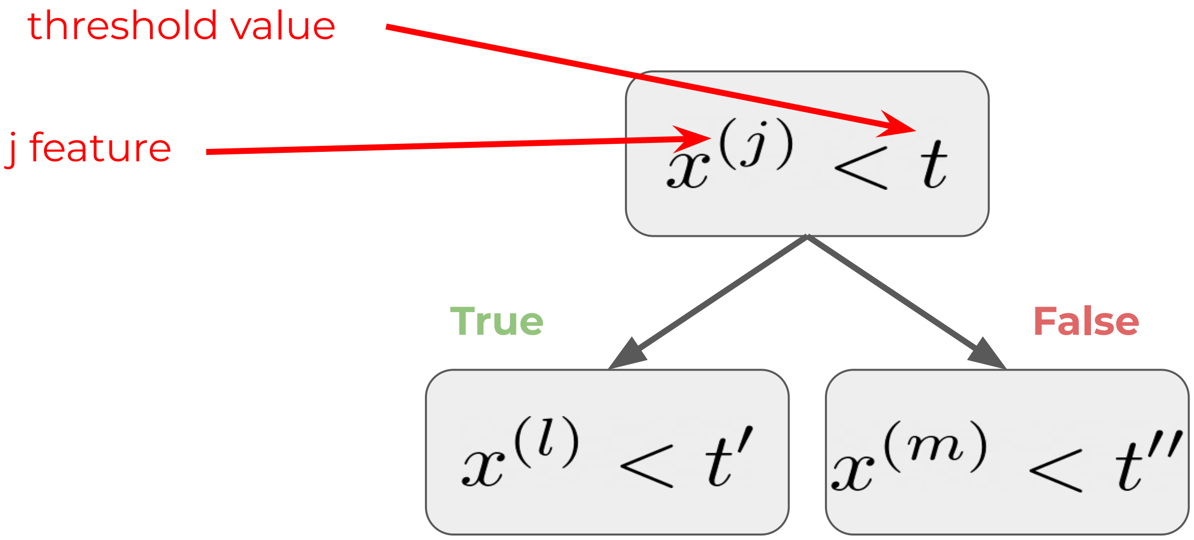


Constructing decision trees



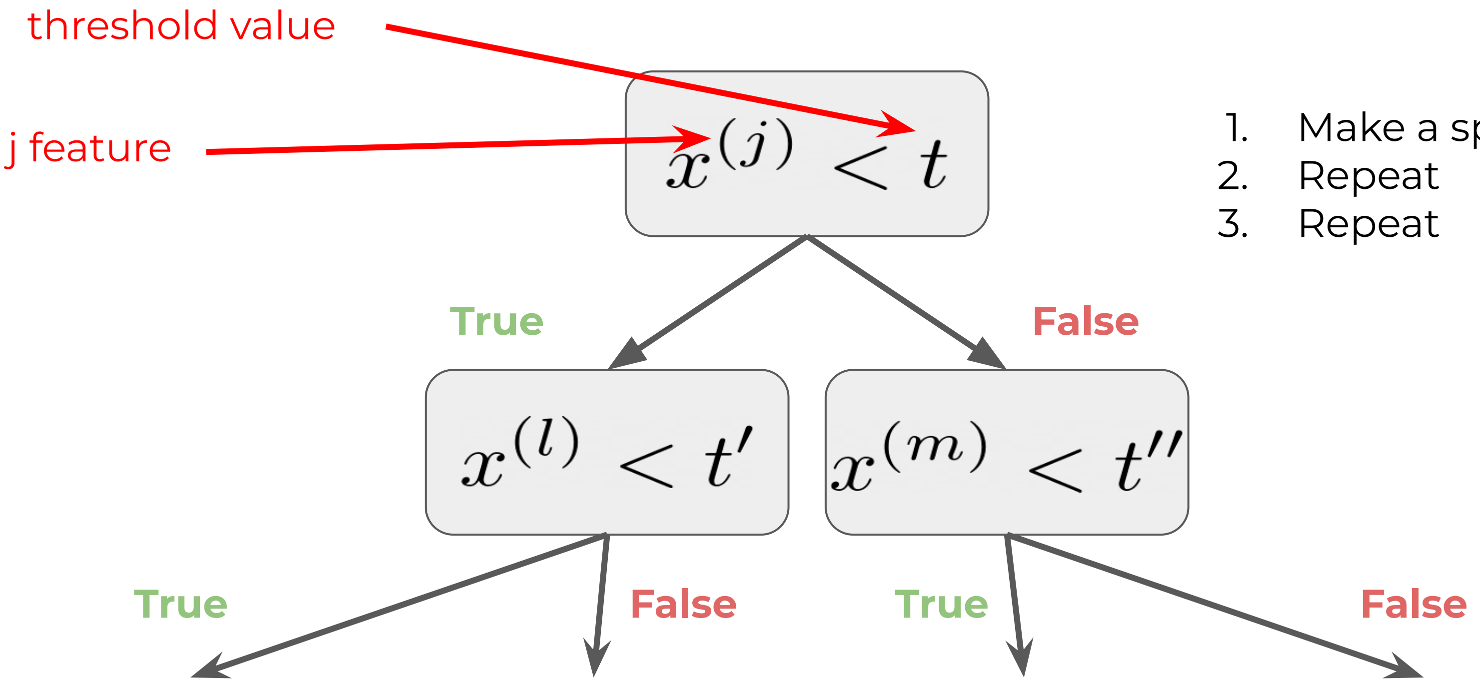
1. Make a split

Constructing decision trees



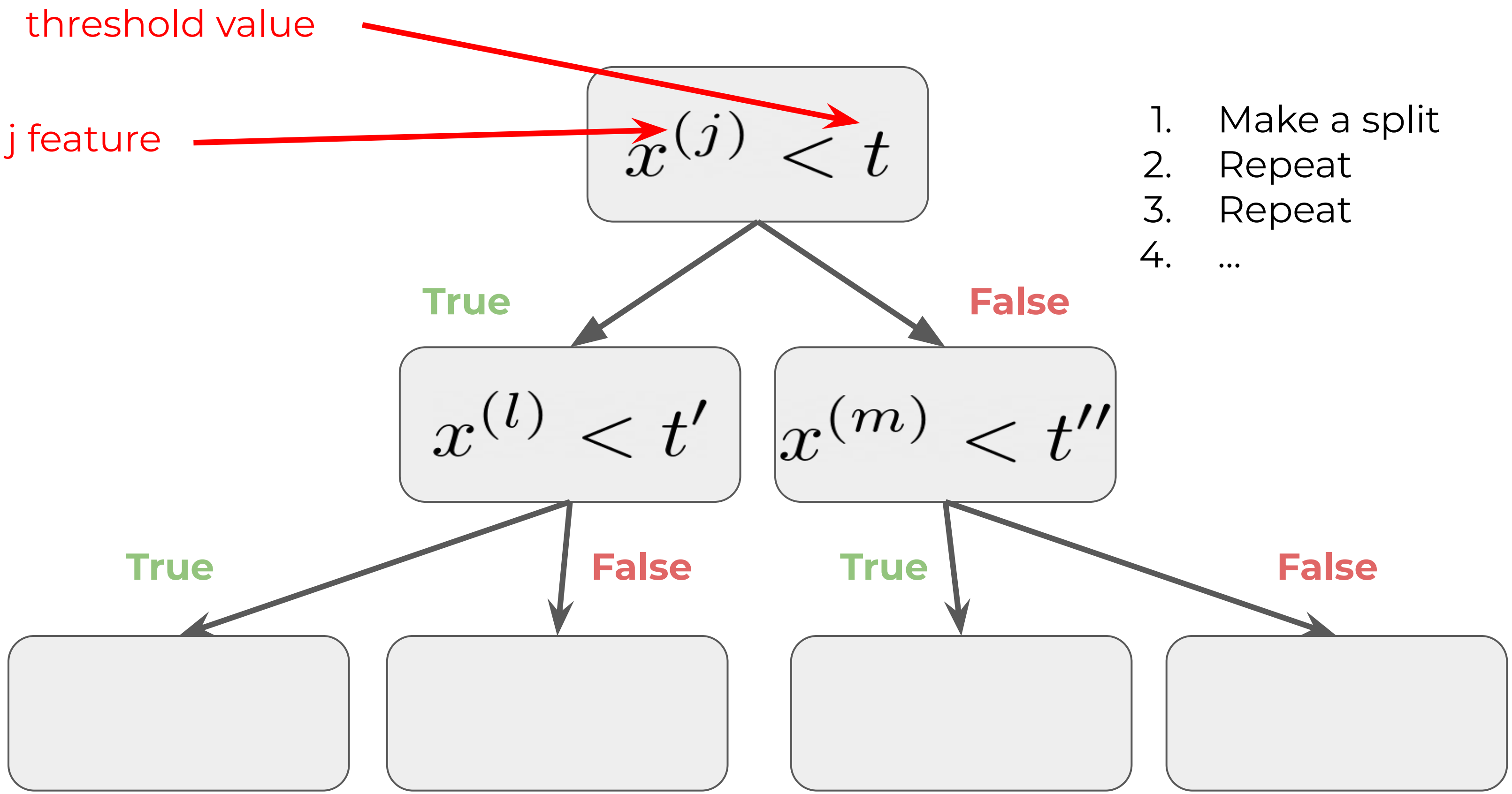
1. Make a split
2. Repeat

Constructing decision trees



1. Make a split
2. Repeat
3. Repeat

Constructing decision trees

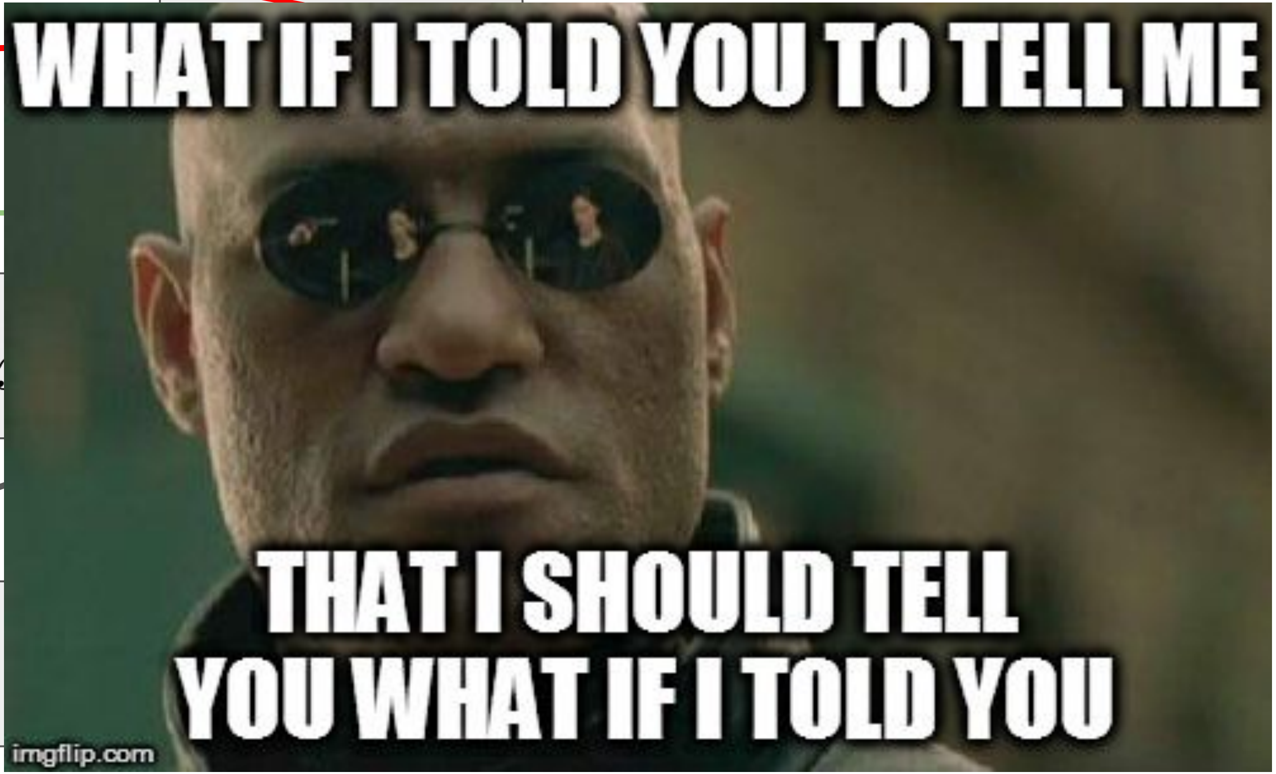


1. Make a split
2. Repeat
3. Repeat
4. ...

threshold value

j feature

True



How to answer in leaf?

Classification:

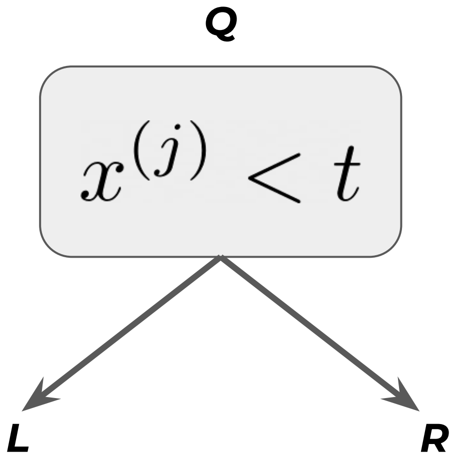
- most popular
- sample with frequencies of classes *(is it good enough?)*

Regression:

Depends on loss function!

- for MSE
    - average in node
- for MAE
    - median in node

How to split data properly?



We can not use gradient this time because solution set is discrete.

So let's apply discrete optimization!

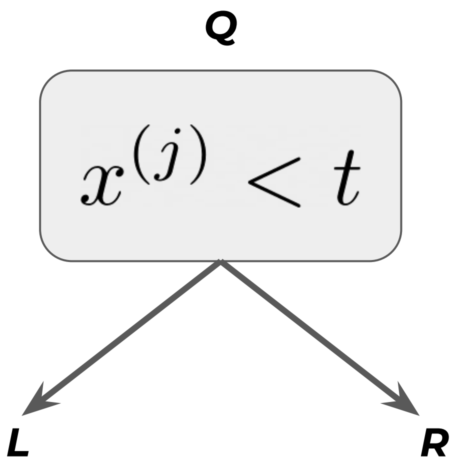
$$\frac{|L|}{|Q|}H(L) + \frac{|R|}{|Q|}H(R) \longrightarrow \min_{j,t}$$

How to choose particular split?

Brute force algorithm will take too much time.

Random splits are chosen and compared.

How to split data properly?



What is H?

$$\frac{|L|}{|Q|} H(L) + \frac{|R|}{|Q|} H(R) \longrightarrow \min_{j,t}$$

Information criteria

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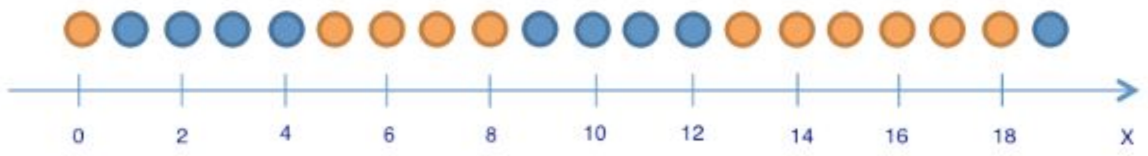
03

Information criteria

H(R) is measure of “heterogeneity” of our data.

Consider binary classification problem:



Information criteria

H(R) is measure of “heterogeneity” of our data.

Consider binary classification problem:

Naive: Misclassification criteria:

$$H(R) = 1 - \max\{p_0, p_1\}$$

1. Entropy criteria:

$$H(R) = -p_0 \log p_0 - p_1 \log p_1$$

2. Gini impurity:

$$H(R) = 1 - p_0^2 - p_1^2 = 2p_0(1 - p_0) = 2p_0p_1$$

Information criteria

H(R) is measure of “heterogeneity” of our data.

Consider multiclass classification problem:

Naive: Misclassification criteria:

$$H(R) = 1 - \max_k \{p_k\}$$

1. Entropy criteria:

$$H(R) = -\sum_{k=0}^{K} p_k \log p_k$$

2. Gini impurity:

$$H(R) = 1 - \sum_k (p_k)^2$$

Information criteria

H(R) is measure of “heterogeneity” of our data.

Consider multiclass classification problem:

Obvious way: Misclassification criteria:

$$H(R) = 1 - \max_k \{p_k\}$$

1. Entropy criteria:

$$H(R) = -\sum_k p_k \log_2 p_k$$

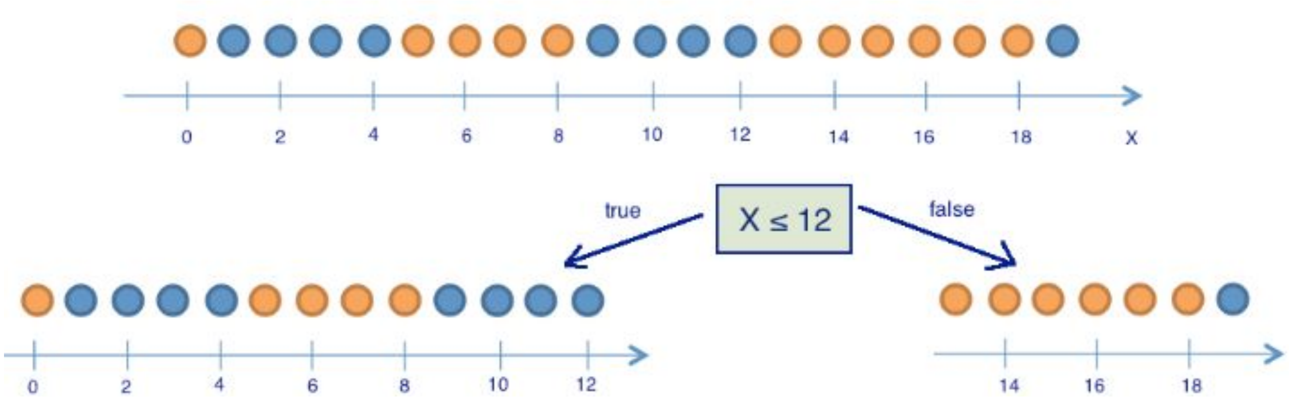
2. Gini impurity:

$$H(R) = 1 - \sum_k (p_k)^2$$

Information criteria

H(R) is measure of “heterogeneity” of our data.

Consider binary classification problem:



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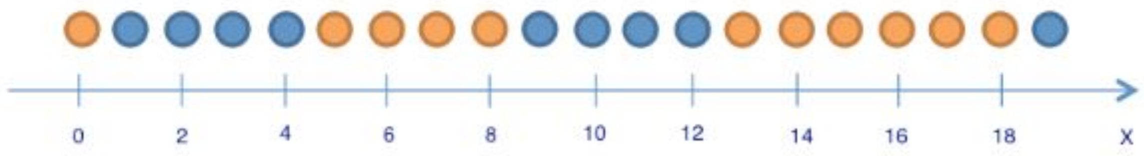

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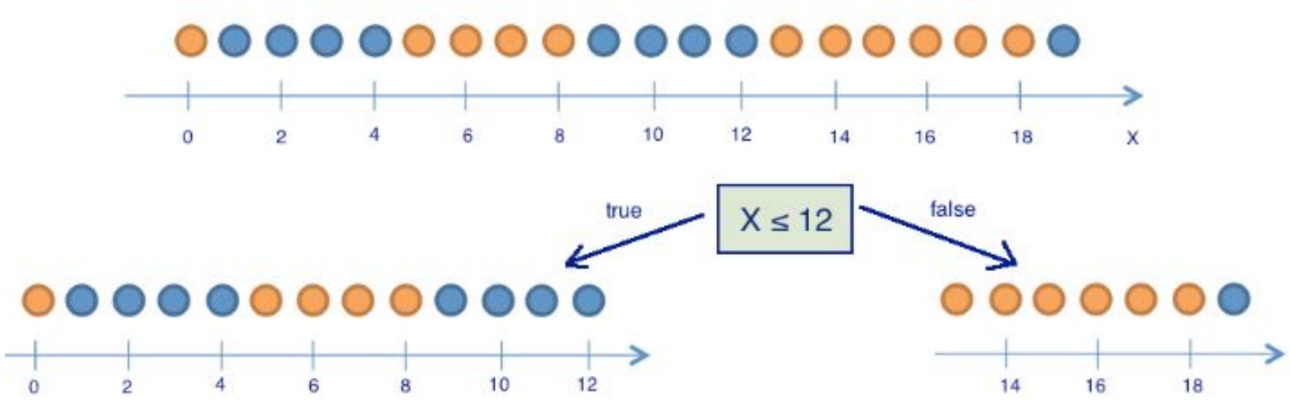
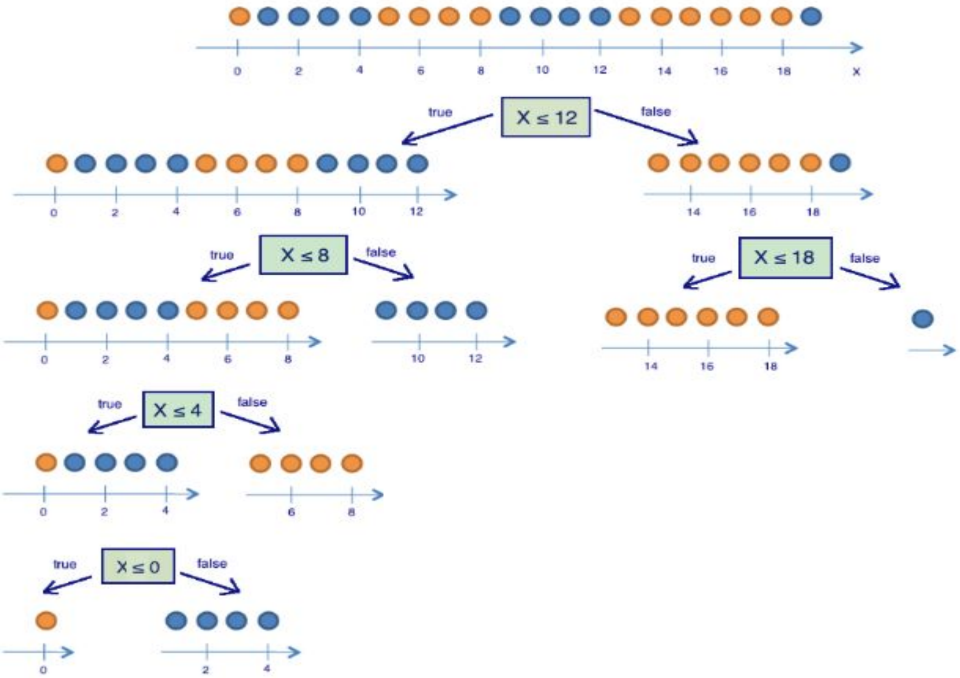


Information criteria: Entropy



$$S = -M\sum_{k=0}^{K} p_k \log p_k$$

In binary case N = 2

$$S = -p_+ \log_2 p_+ - p_- \log_2 p_- = -p_+ \log_2 p_+ - (1 - p_+) \log_2 (1 - p_+)$$

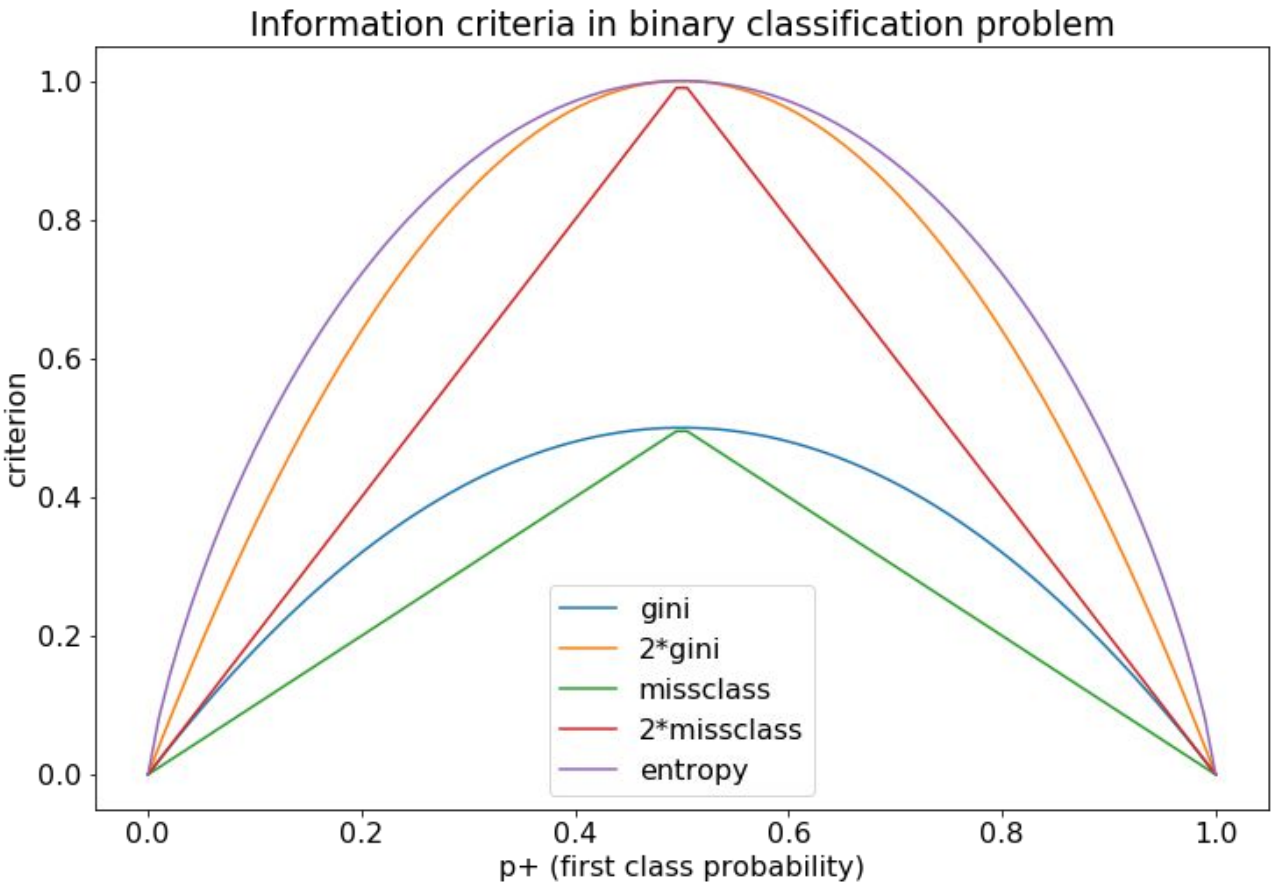
Information criteria: Gini impurity

$$G = 1 - \sum_k (p_k)^2$$

In binary case N = 2

$$G = 1 - p_+^2 - p_-^2 = 1 - p_+^2 - (1 - p_+)^2 = 2p_+(1 - p_+)$$

Information criteria



Information criteria

H(R) is measure of “heterogeneity” of our data.

Consider regression problem:

1. Mean squared error

$$H(R) = \min_c \frac{1}{|R|} \sum_{(x_i, y_i) \in R} (y_i - c)^2$$

What is the constant?

$$c^* = \frac{1}{|R|} \sum_{y_i \in R} y_i$$

Parameters

Variable number of parameters!

Hyperparameters

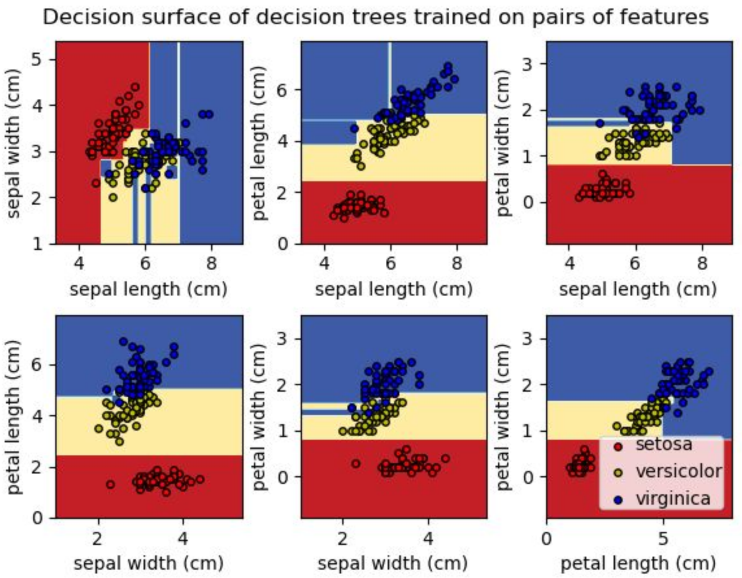
- max_depth: min 1
- min_samples_split: min 2
- min_samples_leaf: min 1
- min_impurity_decrease

Minor

- criterion:
  - gini, entropy for classification
  - MSE or MAE for regression
- splitter: best, random
- max_features: sqrt, log2

As of [sklearn implementation]()

Such hparams could be named pre-pruning or inductive bias



Special highlights

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04

Decision Trees as Linear models

Let J be the subspace of the original feature space, corresponding to the leaf of the tree.

Prediction takes form

$$\hat{y} = \sum_j w_j [x \in J_j]$$

Normalization in trees

Trees are not affected by denormalized data because they choose volume spaces regardless of the scale of both features and target

$$\hat{y} = \sum_j w_j [x \in J_j]$$

Decision Trees as Linear models

Let J be the subspace of the original feature space, corresponding to the leaf of the tree.

Prediction takes form

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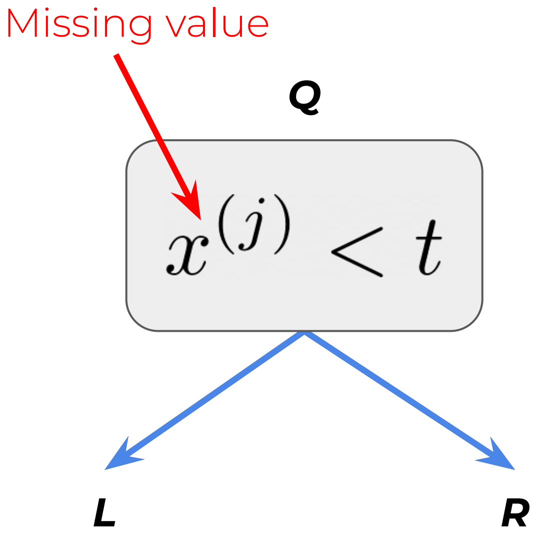
Missing values in Decision Trees

If the value is missing, one might use both sub-trees and average their predictions

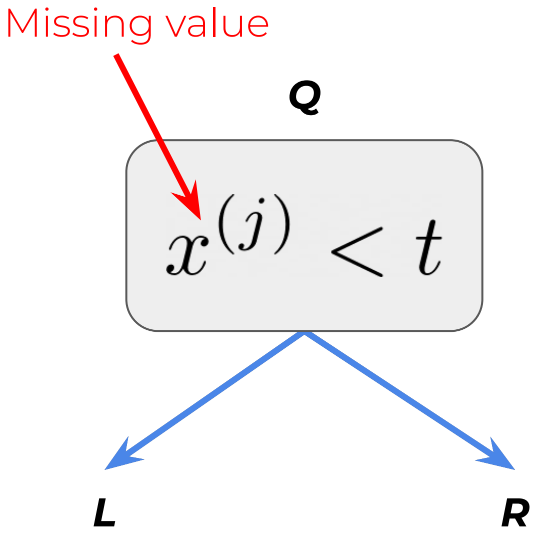


$$\hat{y} = \frac{|L|}{|Q|}\hat{y}_L + \frac{|R|}{|Q|}\hat{y}_R$$

Missing values in Decision Trees

In C4.5 if the value is missing, one might use both sub-trees and average their predictions.

But this will negatively affect model computational performance.



$$\hat{y} = \frac{|L|}{|Q|}\hat{y}_L + \frac{|R|}{|Q|}\hat{y}_R$$

Missing values in Catboost

Forbidden: Missing values are not supported, their presence is interpreted as an error

Min: Missing values are processed as the minimum value (less than all other values) for the feature. It is guaranteed that a split that separates missing values from all other values is considered when selecting trees.

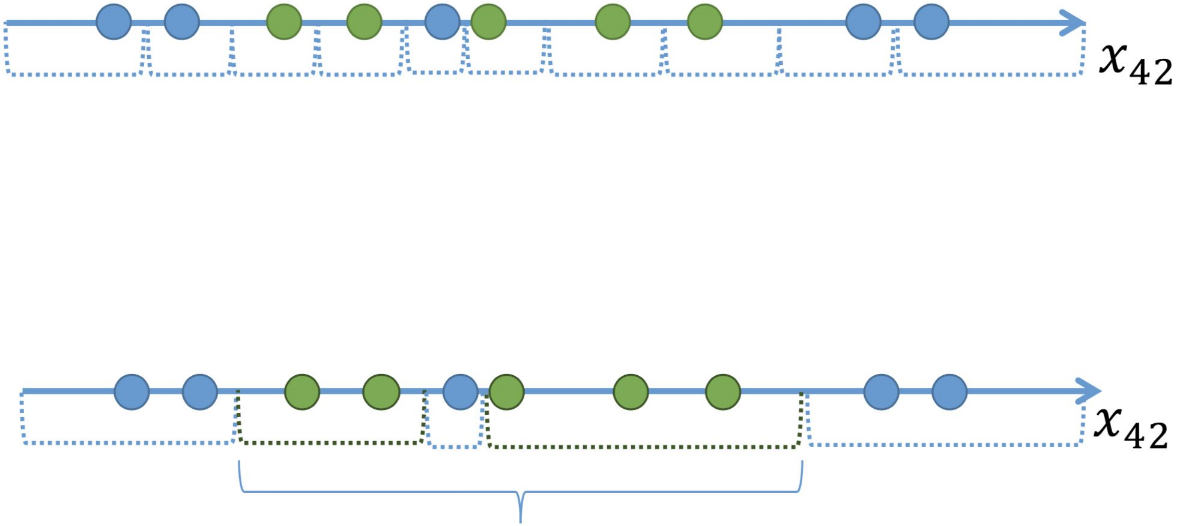
Max: Missing values are processed as the maximum value (greater than all other values) for the feature. It is guaranteed that a split that separates missing values from all other values is considered when selecting trees.

The **default** processing mode **is Min**

Documentation

Binarization

Idea: instead selecting one threshold define several for one feature.



e.g. [Border count hyperparameter]() in Catboost (defaults to 254)

Bootstrap and Bagging

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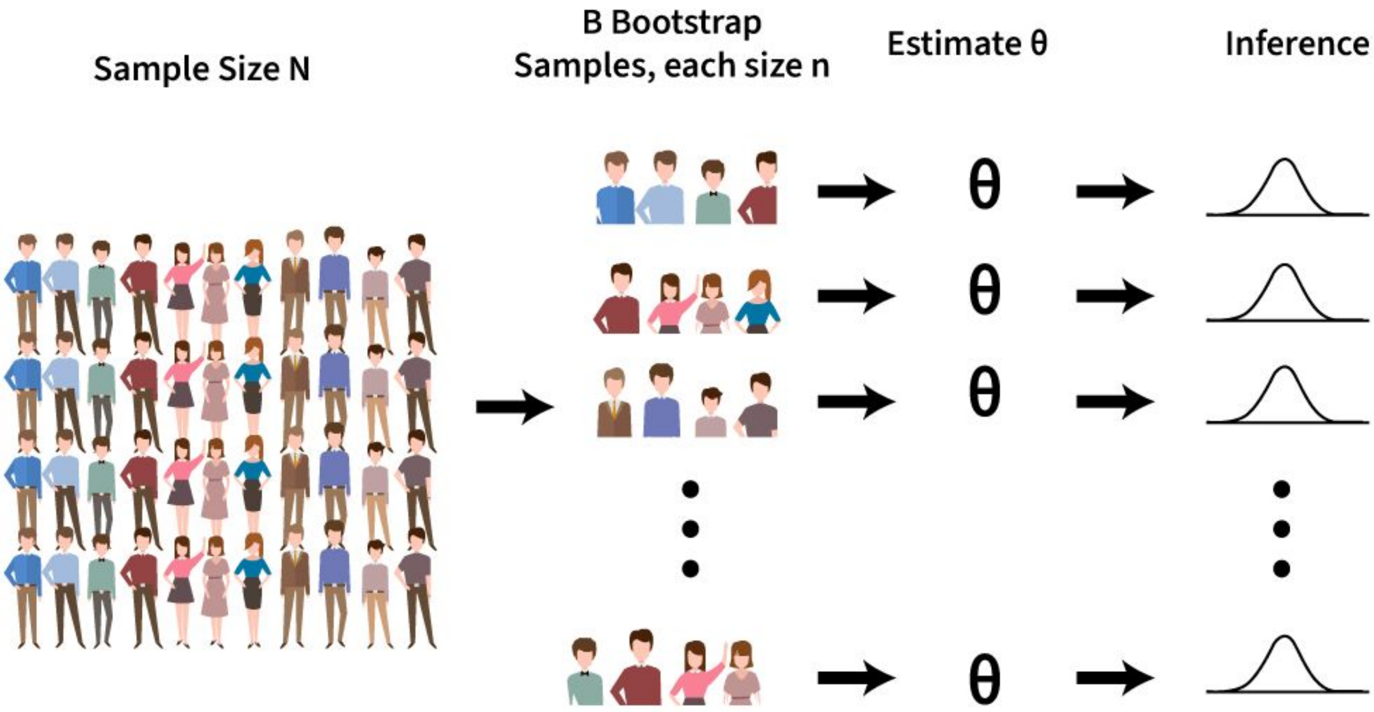
05

Bootstrap



https://www.youtube.com/watch?v=Xz0x-8-cgaQ

Bootstrap

Consider dataset X containing m objects.

Pick m objects with return from X and repeat in N times to get N datasets.

Error of model trained on Xj: $$\varepsilon_j(x) = b_j(x) - y(x), \qquad j = 1, \dots, N,$$

Then $$\mathbb{E}_x(b_j(x) - y(x))^2 = \mathbb{E}_x \varepsilon_j^2(x).$$

The mean error of N models: $$E_1 = \frac{1}{N} \sum_{j=1}^{N} \mathbb{E}_x \varepsilon_j^2(x).$$

Bootstrap

Consider the errors unbiased and uncorrelated:

$$\mathbb{E}_x \varepsilon_j(x) = 0;$$
$$\mathbb{E}_x \varepsilon_i(x)\varepsilon_j(x) = 0, \quad i \neq j.$$

The final model averages all predictions:

$$a(x) = \frac{1}{N}\sum_{j=1}^{N} b_j(x).$$

$$E_N = \mathbb{E}_x\left(\frac{1}{N}\sum_{j=1}^{n} b_j(x) - y(x)\right)^2 =$$
$$= \mathbb{E}_x\left(\frac{1}{N}\sum_{j=1}^{N} \varepsilon_j(x)\right)^2 =$$
$$= \frac{1}{N^2}\mathbb{E}_x\left(\sum_{j=1}^{N} \varepsilon_j^2(x) + \underbrace{\sum_{i \neq j} \varepsilon_i(x)\varepsilon_j(x)}_{=0}\right) =$$
$$\boxed{= \frac{1}{N}E_1.}$$

Error decreased by N times!

Bootstrap

Consider the errors ~~unbiased and uncorrelated~~:

This is a lie

$$\mathbb{E}_x \varepsilon_j(x) = 0;$$
$$\mathbb{E}_x \varepsilon_i(x)\varepsilon_j(x) = 0, \quad i \neq j.$$

$$E_N = \mathbb{E}_x \left(\frac{1}{N} \sum_{j=1}^{n} b_j(x) - y(x) \right)^2 =$$
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$$= \frac{1}{N^2} \mathbb{E}_x \left(\sum_{j=1}^{N} \varepsilon_j^2(x) + \underbrace{\sum_{i \neq j} \varepsilon_i(x)\varepsilon_j(x)}_{=0} \right) =$$
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The final model averages all predictions:

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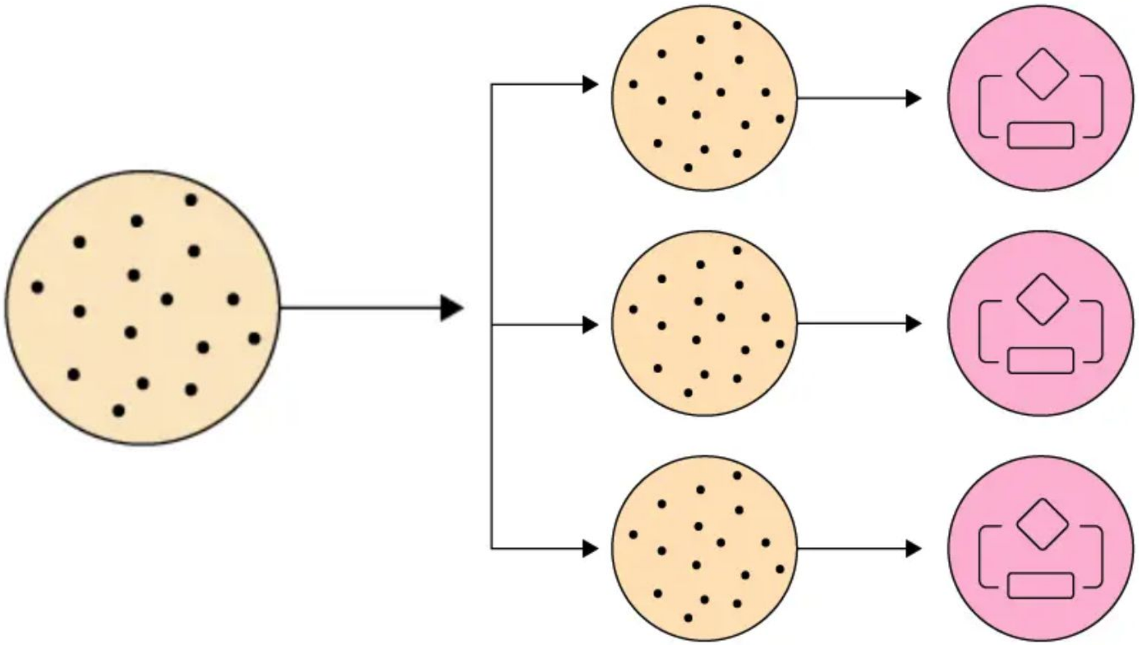
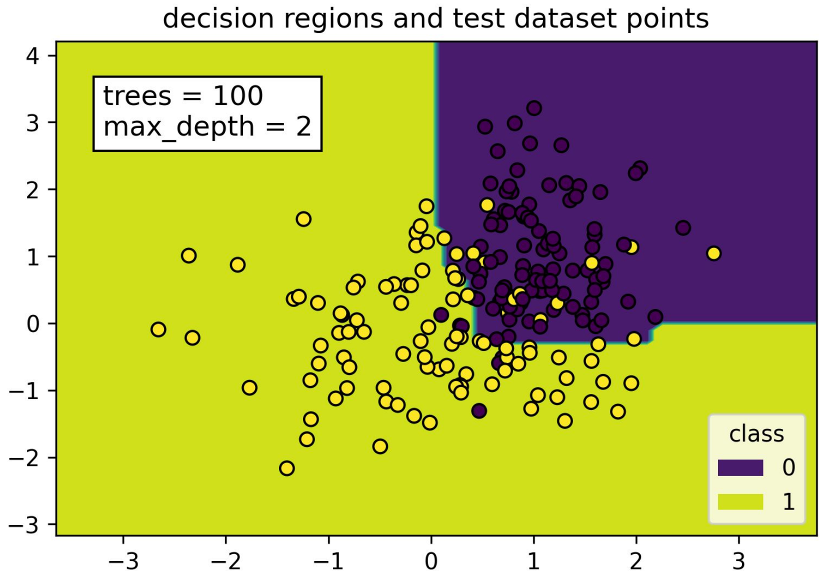
Error decreased by N times!

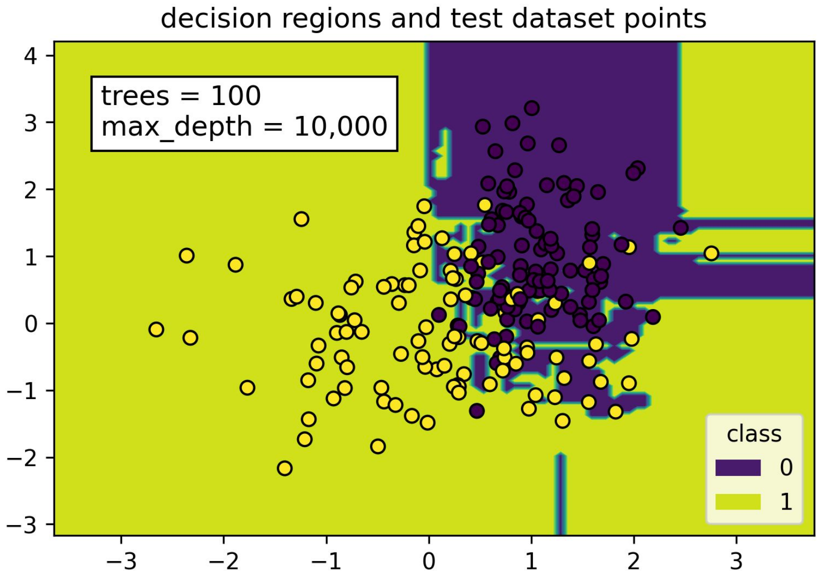
Bagging = Bootstrap aggregating

Decreases the variance if the basic algorithms are not correlated



Bagging overfitting





https://www.tommykhoo.com/machine-learning/rf_overfit/

Random Forest

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06

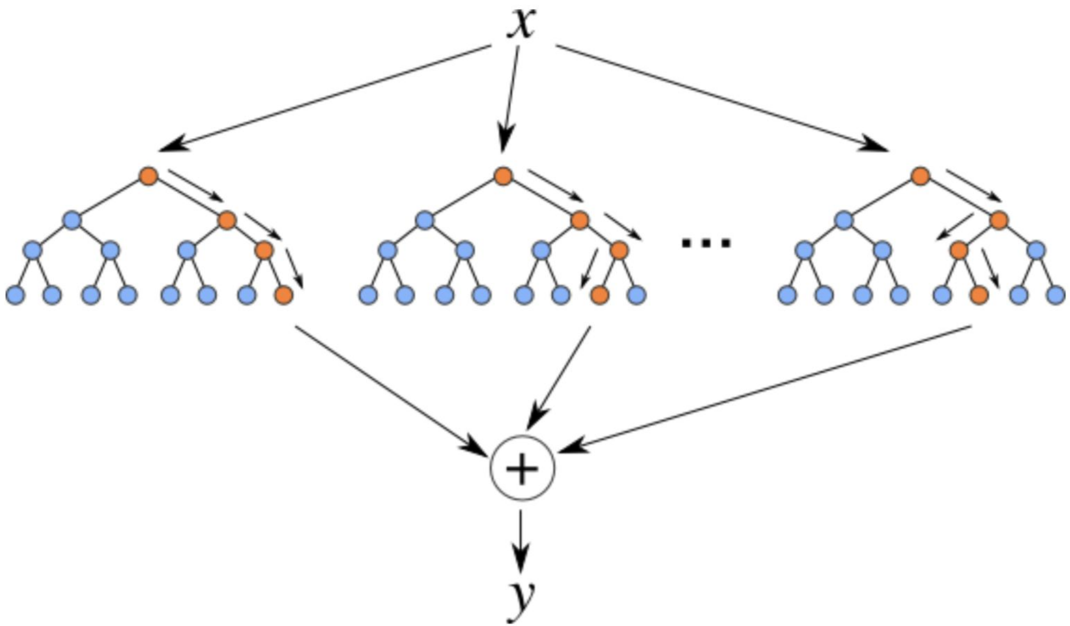
RSM - Random Subspace Method

- Same approach, but with features

- Just subsample features for each bootstrapped dataset

Random Forest

Bagging + RSM = Random Forest



Random Forest

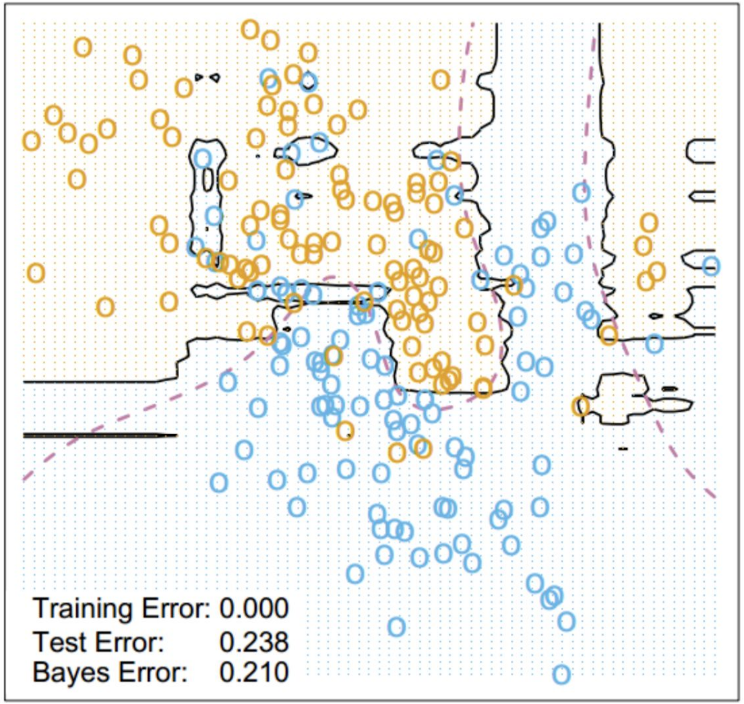
- One of the greatest "universal" models
- Induces additional Variance in data to suppress it then with trees
- There are some modifications:
    - Extremely Randomized Trees
    - Isolation Forest
    - etc.

Random Forest

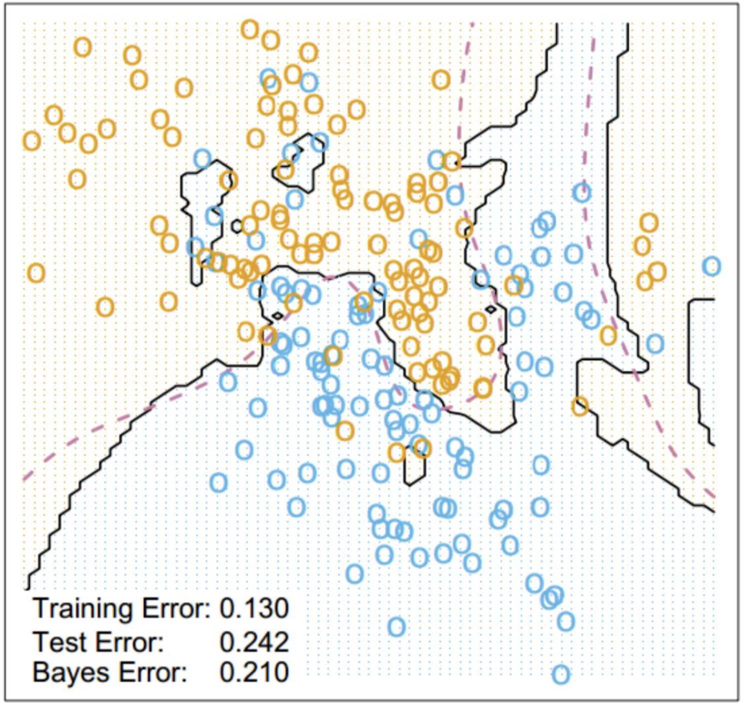
- Allows to use train data for validation: OOB

$$\mathrm{OOB} = \sum_{i=1}^{\ell} L\left(y_i, \frac{1}{\sum_{n=1}^{N}[x_i \notin X_n]} \sum_{n=1}^{N} [x_i \notin X_n] b_n(x_i)\right)$$

Random Forest Classifier



3−Nearest Neighbors



Revise

1. Decision tree: intuition
2. Decision tree construction procedure
3. Information criteria
4. Pruning
5. Decision trees special highlights
   - Decision tree as linear model
   - Dealing with missing data
   - Categorical features
6. Bootstrap and Bagging
7. Random Forest

Thanks for attention!

Questions?

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